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Transformation and migration of Hg in a polluted alkaline paddy soil during flooding and drainage processes☆

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ABSTRACT

Mercury (Hg) contamination in paddy soils poses a health risk to rice consumers and the environmental behavior of Hg determines its toxicity. Thus, the variations of Hg speciation are worthy of exploring. In this study, microcosm and pot experiments were conducted to elucidate Hg transformation, methylation, bioaccumulation, and risk coupled with biogeochemical cycling of key elements in a Hg-polluted alkaline paddy soil. In microcosm and pot experiments, organic- and sulfide-bound and residual Hg accounted for more than 98% of total Hg, and total contents of dissolved, exchangeable, specifically adsorbed, and fulvic acid-bound Hg were less than 2% of total Hg, indicating a low mobility and environmental risk of Hg. The decrease of pH aroused from Fe(III), SO_4^{2-}, and NO_3^- reduction promoted Hg mobility, whereas the increase of pH caused by Fe(II), S^{2-}, and NH_4^+ oxidation reduced available Hg contents. Moreover, Fe-bearing minerals reduction and organic matter consumption promoted Hg mobility, whereas the produced HgS and Fe(II) oxidation increased Hg stability. During flooding, a fraction of inorganic Hg (IHg) could be transported into methylmercury (MeHg), and during drainage, MeHg would be converted back into IHg. After planting rice in an alkaline paddy soil, available Hg was below 0.3 mg kg^{-1}. During rice growth, a portion of available Hg transport from paddy soil to rice, promoting Hg accumulation in rice grains. After rice ripening, IHg levels in rice tissues followed the trend: root > leaf > stem > grain, and IHg content in rice grain exceed 0.02 mg kg^{-1}, but MeHg content in rice grain meets daily intake limit (37.45 μg kg^{-1}). These results provide a basis for assessing the environmental risks and developing remediation strategies for Hg-contaminated redox-changing paddy fields as well as guaranteeing the safe production of rice grains.

1. Introduction

Owing to its ability to transform to toxic and bio-accumulative methylmercury (MeHg) in hydroelectric reservoirs, wetlands, and paddy soils (Li et al., 2009; Xu et al., 2018), Hg threat to ecological environment and human health has raised huge concerns (Wu et al., 2023). As a typical system, paddy fields play an important role in human MeHg exposure (Wang et al., 2021a; Zhao et al., 2016). The elevated MeHg level in rice grains is considered to be tightly associated with inorganic Hg (IHg) methylation (Li et al., 2022a; Li et al., 2019). Paddy soil is a hotspot for MeHg formation (Meng et al., 2024; Wu et al., 2023), since flooding conditions could enhance bacterial metabolism, which are responsible for bacteria-mediated MeHg production (Chen et al., 2019; Peterson et al., 2023). Accordingly, rice plant, could accumulate MeHg concentrations, arousing food safety concerns (Li et al., 2010). Though most studies have investigated the bulk concentration changes in Hg species and interconversion reactions of Hg species (Hellal et al., 2015; Isaure et al., 2020; Liu et al., 2019; Sun et al., 2022; Xu et al., 2019; Zhang et al., 2023b; Zhu et al., 2015), considering the complexity of soil environment, the migration and transformation of Hg is very complicated and affected by numerous environmental factors (Hoggarth et al., 2015; Li et al., 2019; Wang et al., 2021b). Compared with other

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heavy metals, Hg is volatile and methylated, which makes it very challenging to study the dynamics of Hg, and limited our understanding the changes of IHg and MeHg species. Therefore, it is necessary to uncover the underlying mechanisms of Hg migration and transformation, which will help alleviate the problem of Hg-polluted rice grain.

Multiple complex reactions, including desorption, adsorption, reduction, oxidation, demethylation, and methylation, occurred in paddy soil, which could be influenced by pH variations aroused from redox reactions of S/N/C/Fe (Zhang et al., 2023b). During flooding, the reduction of Fe minerals, NO_3^-, and SO_4^{2-} resulted in pH increase, promoting Hg adsorption by soil particles in an acidic paddy soil (Hellal et al., 2015; Zhang et al., 2023b). Reductive dissolution of Fe minerals also reduces Hg adsorption sites, leading to Hg release (O'Connor et al., 2019). Moreover, during flooding, Fe- and sulfate-reducing bacteria promoted Hg reduction and MeHg formation, and meanwhile, methyl trophic bacteria and methanogens mediated demethylation (Isaure et al., 2020). After entering into drainage, Fe(II), NH_4^+, and S^{2-} were rapidly reoxidized by O_2, resulting in pH decrease, which was not conductive to Hg mobilization (Hellal et al., 2015; Zhang et al., 2023b). Meanwhile, owing to Fe(II) oxidation into secondary Fe minerals, a portion of Hg could be re-adsorbed by Fe-bearing particles (Charlet et al., 2002). Generally, in field conditions, the pH of paddy soil range from 5.1 to 8.1 (Santos et al., 2017), and alkaline soils are very common in semiarid and arid climates cover more than 25% of earth's surface (López-Bucio et al., 2000). Though biogeochemical of Hg in an acidic paddy soil has been clarified, it is still lacking a comprehensive understanding of Hg transformation affected by the changes of paddy soil properties. Thus, understanding the transformation and migration of Hg is necessary to accurately predict the fate of Hg in an alkaline paddy soil.

Biogeochemical reactions have been found in the context of heavy metals migration (Chen et al., 2022; Kong et al., 2023), reductive dissolution of Fe minerals (Moore et al., 2012), and surface complexation (Liu et al., 2015). Numerous factors may impact biogeochemical process, including intrinsic factors such as reactive surface area (White and Brantley, 2003), limitation of reactive surfaces in micropore spaces (Kidder and Buchanan, 2008), and extrinsic factors such as heterogeneous distributions of reactive reactant, flow velocity, and reaction space (Enya et al., 2020; Kong et al., 2023; Li et al., 2011). Considering the occurrence of coupled reactions, it is urgent for us to verify whether the mechanism obtained at microscale is applicable to macroscale (Zhu et al., 2023). Various pathways, such as batch, microcosm, column, pot, and field experiments, have been used to investigate the fate and ecological effects of toxic metals, which provide a useful guide for work within a specific condition (Tolle et al., 1985). Generally, pot and field investigations leading to long duration and low tractability of experiments, which are realistic and usually considered superior to microcosm experiments. Whereas microscopic processes are tractable and maintained under defined conditions and usually happen on the smallest possible scale, which is designed to explore the mechanisms under controlled conditions. A pot experiment was a bit more complicated to verify the mechanisms in a simulated environment. Consequently, a combined pathway, including microcosm and pot experiments, is essential for better understanding the migration and transformation of Hg in an alkaline paddy soil.

To resolve these issues, microcosm and pot experiments, using a Hg-polluted alkaline paddy soil, were conducted with a flooding period followed by a sequential drainage to simulate alternating redox conditions. The purposes were to (i) elucidate the underlying mechanism of Hg mobility and methylation in an alkaline Hg-polluted paddy soil in microcosm experiment under controlled conditions, (ii) verify the above proven mechanism in pot experiments with a more complicated environment, and (iii) uncover the potential effectiveness and toxicity of Hg during rice growing in an Hg-contaminated alkaline paddy soil. The findings of this study advance our understanding of Hg dynamics in a polluted alkaline paddy soil under fluctuating redox condition and Hg risk during rice growth.

2. Materials and methods

2.1. Collection and characterization of alkaline paddy soil

Paddy soil at depth of 0–20 cm was collected from a Hg-contaminated paddy field in Tongren City (27°32′57.52″N, 109°12′47.38″E), China. To satisfy the exploration of Hg speciation changes in dried paddy soil and to avoid experimental errors caused by soil heterogeneity during weighing process, the collected paddy soil samples were air-dried, homogenized, and passed through a 2 mm sieve to characterize physicochemical properties. Briefly, the pH was analyzed in a 0.01 M $CaCl_2$ solution with a w/v ratio of 1:2.5 by a Metler PH Meter FE28 (Meng et al., 2024) and Eh was direct determined by a Metler PH Meter FE28. The CEC and OM contents were assessed by ammonium acetate method and dichromate methods, respectively (Purnamasari et al., 2021). To quantify total Hg concentrations, soil sample was digested by the mixture of HNO_3–HCl (v/v, 1:3), and to measure the total Fe and Mn contents, soil sample was digested by the mixture of HNO_3–$HClO_4$–HF (v/v/v, 15:2:2). After extraction, total concentrations of Fe and Mn in the supernatant were measured using inductively coupled plasma-optical emission spectroscopy, and total Hg contents in the supernatant was measured using a Beijing Jitian AFS-933 atomic fluorescence spectrometer (Zhang et al., 2023b). To determine soil texture, paddy soil was analyzed by sedimentation method (Makabe et al., 2009).

As summarized in Table 1, the initial pH of paddy soil was 7.90. The CEC was 26.00 cmol kg^{-1}, which was within the range of CEC in tropical and subtropical soils (Fang et al., 2017). The total contents of Fe and Mn were 26.3 and 0.65 g kg^{-1}, respectively, and total OM concentration was 31.5 g kg^{-1}. The percentages of silt, sand, and clay in paddy soil were 41.36%, 29.92%, and 28.76%, respectively, indicating that paddy soil was loam (Takahashi et al., 2020). Initial total Hg content was 51.14 mg kg^{-1} (Table 1). Based on the previous method (Zhang et al., 2023b), eight Hg species could be distinguished (Table S1). Details of extraction and measurement are provided in supporting information (SI). The contents of organic and sulfides bound Hg and residual Hg accounted for 54.45% and 44.25% of total Hg, respectively, and the proportions of HA bound Hg and Fe–Mn oxides bound Hg were accounted for 1.2% and 0.07% of total Hg, respectively (Table S2). The percentage of available Hg (defined as the sum of F0, F1, F2, and F3) was merely 0.03% (Table S2). The quantification results indicating that Hg has low mobility and was relatively stable in a Hg-polluted alkaline paddy soil (Zhao et al., 2016).

2.2. Microcosm experiments and sample analysis

Microcosm experiments were conducted under alternating redox conditions, in which anoxic and oxic conditions could be well defined and easily regulated (Carpenter, 1996). Microcosm experiments were conducted in 50 mL serum bottles containing 6 g dry paddy soil and 30

Table 1
Physicochemical properties of a Hg-contaminated alkaline paddy soil.

Parameters	Values	Units
pH	7.90 (0.18)	/
CEC	26.00 (0.56)	(cmol kg^{-1})
OM	31.5 (0.67)	(g kg^{-1})
CO_3^{2-}	0.24 (0.011)	(g kg^{-1})
Clay	28.76 (0.84)	(%)
Silt	41.32 (1.02)	(%)
Sand	29.92 (0.84)	(%)
SSA	10.2×10^3 (234.56)	(m^2 kg^{-1})
Total Fe	26.30 (0.73)	(g kg^{-1})
Total Mn	0.65 (0.012)	(g kg^{-1})
Total Hg	51.14 (0.78)	(mg kg^{-1})

Abbreviations of OM, CEC, and SSA represent organic matter, cation exchange capacity, and specific surface area, respectively.

mL 5 mM NaCl. Soil suspension were flushed with nitrogen for 30 min and sealed with butyl rubber stoppers (Liu et al., 2015). Controlling soil redox conditions in a constantly shaken microcosm system has several advantages over static incubations. That is, redox conditions are well defined and reproducible, and can be rapidly changed. In addition, the impact of E_H can be investigated independently from other soil parameters (Rinklebe et al., 2016; Rupp et al., 2010). Thus, a total of 30 serum bottles were placed in a horizontal shaker and shake with a speed of 220 rpm min^{-1} at 25 °C in the dark. After 40 d of flooding, all butyl rubber stoppers were replaced with a 0.45 μm porous film to allow gas exchange with air for 20 d of drainage under continuous shaking (Hong et al., 2023).

At flooding intervals (0, 2, 5, 10, 20, and 40 d) and drainage intervals (2, 7, 15, and 20 d), three serum bottles were obtained and crimp-sealed cap was removed in an anaerobic chamber (Plas-Labs, USA, H_2/N_2 (2/98, v/v)), and all sampling processes are destructive. Microcosmic incubation experiments simulated the occurrence of representative flooding/drainage paddy soils, following the method described by previous study (Wang et al., 2019). Because drainage usually occurs 2–3 weeks before rice maturity, flooding and drainage periods were set to 40 and 15 d, respectively. The Eh and pH were immediately determined by Eh and pH meter, respectively. Then, remaining suspension samples were transferred into 50 mL centrifuge tubes and centrifuged at 5000 rpm for 15 min. After centrifugation, supernatants were filtered by 0.45 μm nylon filters. Obtained filtrate was used for water-soluble Hg (F0), water-soluble MeHg (WMeHg), dissolved organic carbon (DOC), NH_4^+, NO_3^-, Fe(II), and SO_4^{2-} analyses. The residual wet paddy soil samples in centrifuge tubes were used to determine the contents of extractable Fe (II), IHg, total MeHg (TMeHg), and effective MeHg (EMeHg).

Aqueous NO_3^- and SO_4^{2-} contents were determined by ion chromatography (ICS-600, Dionex, U.S.A.), and NH_4^+ concentration was analyzed by a continuous flow analyzer (Hu et al., 2022a). DOC contents in the supernatant were measured using a total organic carbon analyzer (Shimadzu, Japan) (Hu et al., 2020; Hu et al., 2022b). The 0.5 M HCl was selected to determine the extractable Fe(II) (defined as $Fe(II)_{HCl}$), and after extraction, dissolved Fe(II) and $Fe(II)_{HCl}$ concentrations were immediately (Hu et al., 2021a; Hu et al., 2021b). For IHg species, TMeHg, and EMeHg determination, suspension samples were first acidified with 5% HCl (Liu et al., 2022a; Zhang et al., 2023b). Collected suspension samples were determined by a modified Tessier SEP to determine IHg species (Anju and Banerjee, 2010). The filtrate was distilled to concentrate for determining WMeHg (Dos Santos et al., 2009). Wet paddy soil was air-dried in an anaerobic chamber and dried paddy soil was collected for TMeHg and EMeHg analyses. Specially, a portion of dried paddy soil was used to sequential extraction with 20% $CuSO_4$–HNO_3 and CH_2Cl and water back-extraction with heating to quantify TMeHg (Hight and Cheng, 2006). The remaining portion was extracted by $(NH_4)_2S_2O_3$ to quantify EMeHg, which was considered as bioavailable MeHg (Wang et al., 2023b). All MeHg extractions were ethylated into methylethyl-Hg for determination by MERX cold vapor atomic fluorescence spectroscopy (Brooks Rand Instruments, U.S.A.) based on US EPA method 1630. Standards ERM-cc580 (estuarine sediment) and GBW07453 (GSS-24) for MeHg and THg analyses, respectively, were selected for quality control. Details are provided in SI.

2.3. Pot experiments and sample analysis

The seeds of rice *Oryza sativa* L. from cultivar Huanghuazhan were first sterilized in 10% H_2O_2 solution for 10 min and then flashed with deionized water (>18.2 MΩ) for 30 times (Zhong et al., 2021). To ensure moistened seeds germinate, all seeds were placed in a biochemical incubator and incubated at 25 °C. After germination, all seedlings were placed on seedling-raising plates with unpolluted soil in a climate-regulated greenhouse. After two weeks of growth, all rice seedlings were transplanted into 21 experimental pots filled with 6 kg of Hg-polluted paddy soil in a climate-regulated greenhouse at 25 °C. Each experimental pot was planted with 12 rice seedlings. To keep the water level about 5 cm above soil surface during flooding and to keep a 75% soil moisture content during drainage process, deionized water was added to experimental plots daily (Liu et al., 2023). To maintain normal growth of rice, urea as N fertilizer and KH_2PO_4 and $K_2HPO_4{\cdot}3H_2O$ as P and K fertilizers were added to each experimental pot at a rate of 0.2145 and 0.0625 g kg^{-1} dry soil, respectively (Zhong et al., 2021).

During rice growth, paddy soil and rice plant samples were collected. Specially, flooding duration was approximately 90 d, and drainage usually occurred in the last 2–3 weeks of filling period before rice maturity (Zhong et al., 2021). Thus, to simulate the real growth process of rice, sampling points during flooding were set to 0, 15, 30, 60, and 90 d, and sampling points during drainage were set 97 and 105 d. The pH and Eh at a depth of 5 cm were monitored by pH and Eh meter. After pH and Eh analysis, paddy soil solution was obtained by a Rhizon sampler, and during collection, single-use samplers were inserted 15 cm to collect liquid (Kuzyakov et al., 2007). Obtained paddy soil samples were immediately filtered through a 0.22 μm membrane. Wet topsoil at a depth of 5–20 cm was collected and placed into polypropylene centrifuge tubes for Fe and Hg species analyses. During rice growth, rice plants were obtained, washed, and digested for indicators analysis. Note that iron plaque should be washed from roots to avoid the interfering with the analysis of Hg species. Generally, Hg usually present in inorganic and organic forms in paddy soil and rice tissues, and MeHg dominantly was derived from microbes acting on IHg. Since Hg contributes to human persecution via amplification and enrichment in the direct inhalation or food chain. Rice was demonstrated to be one of the dominant pathways of MeHg ingestion by humans, especially in regions where rice is a staple food. To quantify the organic and inorganic Hg in rice, only IHg concentrations in the roots, stems, leaves, and grains of rice plants as well as MeHg concentrations in grains at maturing stage were quantified (Pang et al., 2019). The details of sample analysis are provided in section 2.2.

3. Results and discussion

3.1. Variations in physicochemical properties of paddy soil

Microbial metabolic activities could consume H^+ and produce CO_2 (Kroukamp and Wolfaardt, 2009), which promoted an increase in pH owing to H^+ consumption and a decrease in pH due to the absorption of CO_2 by alkaline paddy soil solution. During 40 d of flooding, the pH and Eh decreased simultaneously with times (Fig. 1a). This implied that CO_2 absorption by soil solution has a greater effect on pH variation. The decrease in pH was mainly due to the oxidation of oxidizing substances (e.g., Fe(III), SO_4^{2-}, and NO_3^-) by Fe and sulfate-reducing bacteria (Beckers et al., 2019a; Beckers et al., 2019b) and organic matter by *Bacillus subtilis* and *Pseudomonas fluorescens* (Das and Chandran, 2011; Davey and O'Toole G, 2000). Owing to the rapid reductive dissolution of Fe minerals and adsorption of produced Fe(II) by paddy soil particles (Lu et al., 2023; Meng et al., 2024), extractable Fe(II) ($Fe(II)_{HCl}$) concentration increased during 5 d of flooding (Fig. 1b). After 5 d of flooding, extractable Fe(II) remained unchanged, which may be because the produced Fe(II) was basically used to synthesize Fe(II)-bearing Fe minerals. Whereas no aqueous Fe(II) was detected (Fig. 1b), which may be ascribed to the fact that alkaline condition was favorable for Fe(II) absorption and Fe(II) precipitation (Chi et al., 2023; Huang et al., 2023). The changes in NO_3^- and SO_4^{2-} were opposite to that of NH_4^+ concentration (Fig. 1c). Specially, in the first 2 d of flooding, NO_3^- concentration dropped rapidly to 0 and then kept constant, and SO_4^{2-} contents gradually decreased to 0 in the first 20 d of flooding and subsequent remained unchanged. The decrease of NO_3^- and SO_4^{2-} concentrations may be due to microbially mediated the reduction of these two substances (Hong et al., 2023; Meng et al., 2024). During flooding, a small amount of NH_4^+ was detected and its concentration increased with time (Fig. 1c), which was derived from NO_3^- reduction (Nie et al., 2021). DOC

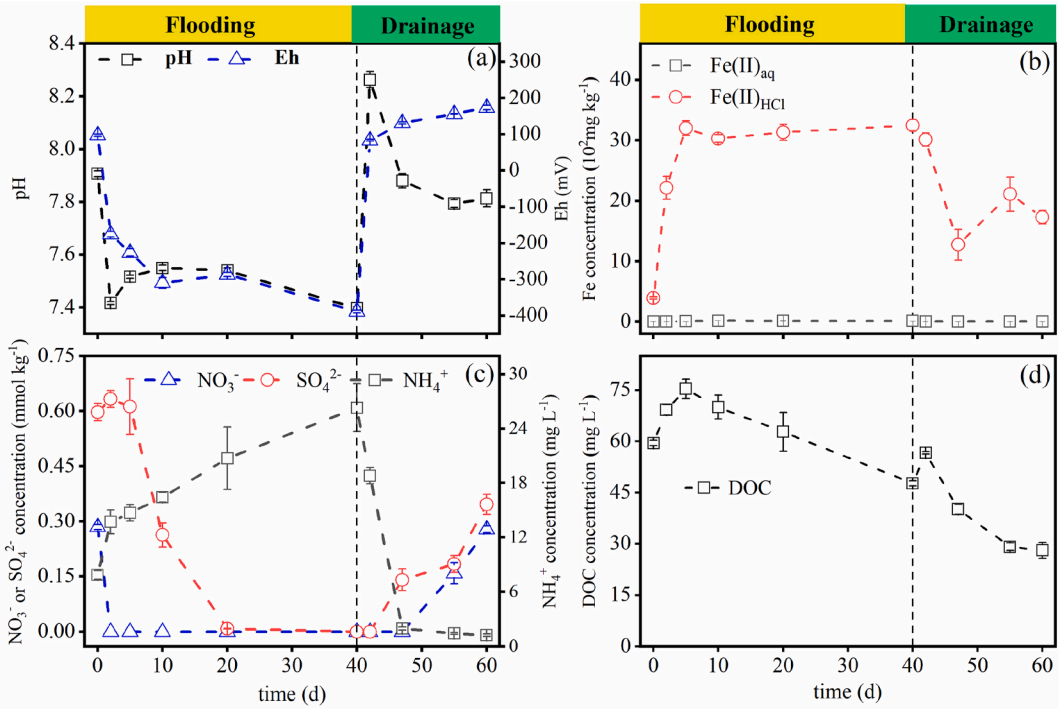


Fig. 1. (a) Paddy soil pH and Eh, (b) dissolved Fe(II) and $Fe(II)_{HCl}$ (c) NO_3^-, SO_4^{2-}, and NH_4^+, and (d) DOC concentrations in the microcosm experiments during 40 d of flooding and 20 d of drainage. Abbreviation of DOC represent dissolved organic carbon.

concentration increased within the first 5 d of flooding and then gradually decreased over time (Fig. 1d), which may be attributed to the release of low molecular OM owing to decomposition by bacteria such as *Bacillus subtilis* and *Pseudomonas fluorescens* (Das and Chandran, 2011; Davey and O'Toole G, 2000), followed by Hg complexation and/or bacterial consumption (Gu et al., 2011).

After entering drainage, the pH first increased rapidly because a large number of H^+ were produced, and then gradually decreased to 7.81 (Fig. 1a) due to the absorption of CO_2 from air by an alkaline paddy soil solution (Usui et al., 2003). Whereas Eh increased with time (Fig. 1a) since the reduced substances, including Fe(II), S^{2-}, and NH_4^+, were oxidized by O_2. Aqueous Fe(II) remained at low level and $Fe(II)_{HCl}$ contents maintained a downward trend (Fig. 1b) because of Fe(II) oxidation by O_2 (Morgan and Lahav, 2007). Owing to the oxidation of S^{2-} by O_2 during drainage (Chen and Morris, 1972) and anaerobic ammonium oxidation (Zhu et al., 2018), NO_3^- and SO_4^{2-} concentrations increased with time and NH_4^+ content decreased to 0 (Fig. 1c). The decrease in DOC was further enhanced by adsorption and/or coprecipitation with newly-formed Fe minerals from Fe(II) oxidation (Fig. 1d).

3.2. Dynamics of Hg speciation during microcosm experiments

The change of soil properties will inevitably affect Hg mobility and transformation, and coupled reactions occurred during flooding and drainage (Hong et al., 2023; Meng et al., 2024; Zhang et al., 2023b). Overall, after 40 d of flooding and 20 d of drainage, residual Hg and organic- and sulfides-bound Hg contents were close to 98% of total Hg (Fig. 2a), indicating that most Hg was fixed. Whereas HA bound Hg increased slightly from 1.62% to 2.64% and Fe–Mn oxides bound Hg was below 0.1% (Fig. 2a). This indicated that Hg was more feasible complexed with sulfide and OM fractions than that with Fe mineral fractions (Zhang et al., 2023b). Organic- and sulfides-bound Hg represents Hg binding with larger molecular weight OM or binding on matrix surfaces, which were usually rich in OM with SH- groups (Khwaja et al., 2006; Liu et al., 2022b). Owing to the covalent combination of Hg and HA functional groups to form gridding chelates ((Khwaja et al., 2006), HA bound Hg was insoluble and unavailable. FA bound Hg was readily available for rice because of FA functional groups association with Hg by electrostatic forces (Pang et al., 2019). This results in the amounts of HA bound Hg being significantly higher than those of FA bound Hg (Fig. 2a).

Soluble and exchangeable IHg and MeHg fractions are potentially good indicators of Hg bio-accessibility and presents highest environmental risk (Huang et al., 2020). Since no significantly change of F0, F1, F2, F3, and F4 were observe in Fig. 2a, and thus, temporal changes of F0, F1, F2, and F3 concentrations were further depicted in Fig. 2b to identify subtle changes. The contents of exchangeable, carbonate bound Hg, and FA bound Hg kept at low level and their concentrations were all less than 0.01 mg kg^{-1} (Fig. 2b). Because the pH was greater than 7.4 (Fig. 1a), which could effectively fix Hg, causing dissolved Hg concentration fluctuating at a low level (Fig. 2b). Specially, at the first 2 d of flooding and 7 d of drainage, a significant increase in dissolved Hg ($p < 0.01$) (Fig. 2b), suggesting that a part of adsorbed Hg was released into solution. This may be ascribed to the reductive dissolution of Fe minerals (Fig. 1b) or HgS oxidization, leading to the release of Fe–Mn oxides bound Hg and organic- and sulfide-bound Hg (Barnett et al., 2001; Hellal et al., 2015; Zhang et al., 2023b). Moreover, a decrease in pH (Fig. 1a) may also slightly facilitated Hg desorption from paddy soil, promoting an increase in dissolved Hg (Fig. 2b). After 2 d of flooding and 7 d of drainage, dissolved Hg decreased rapidly (Fig. 2b), which may be ascribed to the fact that Hg immobilization by newly-formed Fe minerals through adsorption or co-precipitation processes (Zhang et al., 2012; Yang et al., 2021; Zhang et al., 2023b).

The variations in WMeHg, EMeHg, and TMeHg showed different tendencies and the accumulation of MeHg was a dynamic balance between MeHg demethylation and IHg methylation (Zhang et al., 2023b). Notable, during drainage and flooding, TMeHg concentration was higher than the sum of WMeHg and EMeHg concentrations, indicating that a portion of MeHg was existed as residual species (Rand et al., 2015; Yang et al., 2023b). Specially, during flooding, WMeHg content gradually decreased from 0.6 to 0 μg kg^{-1}, whereas EMeHg concentration was below 0.01 μg kg^{-1} and its concentration remained basically unchanged (Fig. 2c). This may be due to the rapid uptake of the produced WMeHg

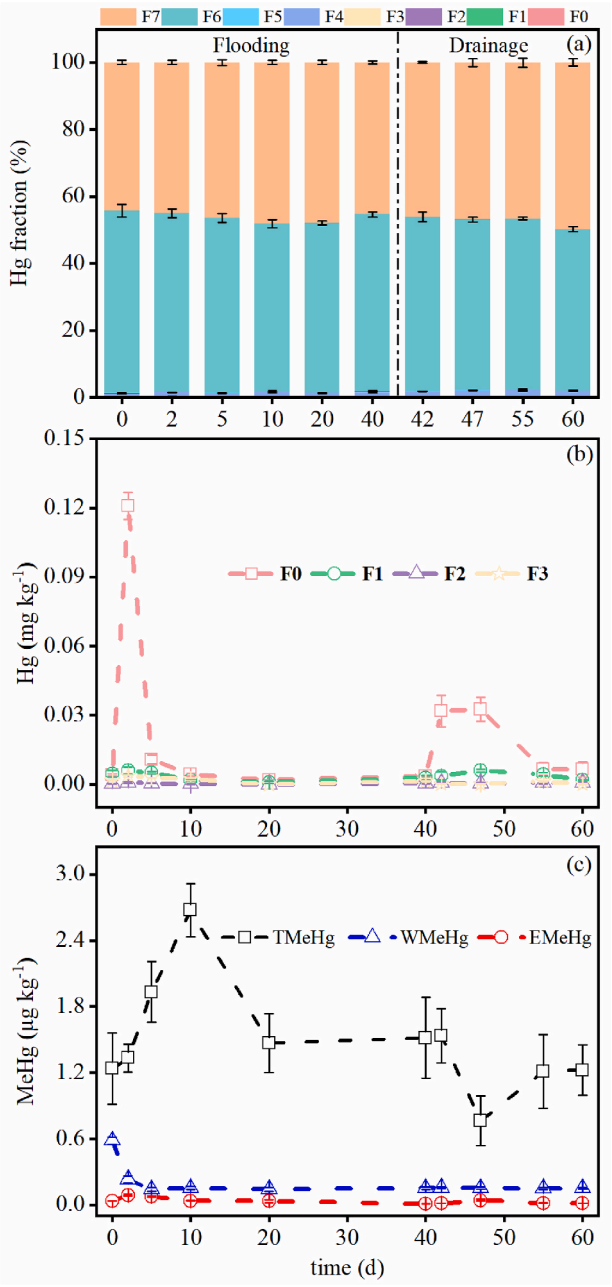


Fig. 2. (a) Hg species, (b) available Hg concentration, and (c) TMeHg (total MeHg), WMeHg (water-soluble MeHg), and EMeHg (effective MeHg) contents. F0 denotes dissolved Hg; F1 denotes exchangeable Hg; F2 denotes carbonates bound Hg; F3 denotes fulvic acid bound Hg; F4 denotes humic acid bound Hg; F5 denotes Fe–Mn oxides bound Hg; F6 denotes organic and sulfides bound Hg; F7 denotes residual Hg.

and EMeHg by rice plants from paddy soil (Palansooriya et al., 2020), leading to WMeHg and EMeHg concentrations remained unchanged. Interestingly, TMeHg content first increased from 1.23 to 2.67 μg kg^{-1} during 10 d of flooding and then decreased to 1.21 μg kg^{-1} (Fig. 2c). This may be due to the rapid methylation of free Hg by Hg methylating bacteria, including methanogens, iron-reducing bacteria, and sulfate-reducing bacteria (Peterson et al., 2023; Tang et al., 2020; Thomas et al., 2018), and demethylation rate may be greater than methylation rate in the process of late flooding period (Hoggarth et al., 2015; Zhang et al., 2023b). During drainage, WMeHg and EMeHg contents was maintained at 0 and TMeHg concentration decreased with drainage time (Fig. 2c). The variation trend of TMeHg might be attributed to the coprecipitation and adsorption of MeHg onto newly-formed Fe minerals from Fe(II) oxidation (Zhang et al., 2023a). The contents of TMeHg during flooding was higher than that in the process of drainage (Fig. 2c). This may be because the available DOC in flooding duration was higher than that in drainage duration, which would promote Hg methylation by bacteria (Li et al., 2022b; Zhang et al., 2023b). Interestingly, on 40 d, there was a significant increase in available Hg (Fig. 2b), whereas TMeHg exhibited a notable decrease (Fig. 2c). This may be ascribed to the fact that oxidative demethylation rate of MeHg may be higher than methylation rate of IHg during oxic process (Lu et al., 2016; Zhao et al., 2020).

3.3. *Geochemical processes of S/N/C/Fe coupled with Hg species variations during pot experiment*

To monitor the variations in element cycling and Hg speciation at paddy soil and water interfaces in pot experiments, relevant samples were characterized. The Eh and pH decreased during 90 d of flooding and increased during 15 d of drainage (Fig. 3a). For Fe(II) species, aqueous Fe(II) was close to detection limit during rice growth, whereas extractable Fe(II) concentration increased with flooding time and decreased with drainage time (Fig. 3b). The contents of NO_3^- and SO_4^{2-} declined rapidly in the first 15 of flooding and its concentration kept at 0 mmol kg^{-1} within the rest of flooding period (Fig. 3c). After entering drainage, NO_3^- and SO_4^{2-} concentrations increased over time (Fig. 3c). Within 15 d of flooding, DOC concentration increased to peak value, and subsequently decreased during remaining flooding and drainage periods (Fig. 3d). Interestingly, DOC concentration in the microcosm experiment was lower than that in the pot experiment (Fig.. 2d and 3d). This may be due to the fact that rice released an amount of root exudates during growth period, causing a significant increase in DOC (Aulakh et al., 2001).

During rice growth, organic-, sulfide-bound, and residual IHg were the dominant Hg species, accounting for 98% of total Hg. HA bound Hg was about 1%, and Fe–Mn oxides bound Hg and available Hg were below 0.01% (Fig. 4a). Specially, after 90 d of flooding, organic and sulfides bound Hg content decreased and residual Hg concentration increased, whereas the content of available IHg, the sum of dissolved, exchangeable, carbonates bound, and FA bound Hg, remined unchanged. After 15 of drainage, residual Hg content decreased and organic and sulfides bound Hg concentration decreased, whereas available Hg content increased firstly and then decreased (Fig. 4a). The variations in available Hg can be attributed to the changes of pH and Eh promoting OM decomposition, oxidative dissolution of HgS, and Hg adsorption by soil minerals (Zhang et al., 2023b). Moreover, the redox of Fe-bearing minerals and reduction of SO_4^{2-} altered the adsorption capacity of soil particles, which affected the adsorption of soil particles to Hg via co-precipitation or adsorption processes (Vasques et al., 2020; Zhao et al., 2016). Available Hg in the pot experiments was below 0.03 mg kg^{-1} (Fig. 4b), indicating that low Hg mobility in an alkaline paddy soil after planting rice. WMeHg contents decreased to 0 μg kg^{-1} after 15 d of flooding and then kept unchanged during remaining period of rice growth, whereas no EMeHg was detected (Fig. 4c). For TMeHg, its contents increased to 4.86 μg kg^{-1} after 30 d of flooding and then gradually decreased with time (Fig. 4c). The increase in MeHg may be ascribed to the production of FeS and FeS_2, promoting Hg methylation (Zhao et al., 2016), whereas the decrease in MeHg may be attributed the production of insoluble HgS (Lei et al., 2021).

3.4. *Variations of Hg speciation in rice plant under the impact of flooding and drainage*

Understanding Hg transformation at the interfaces of paddy soil and rice would help us to know the underlying risk of Hg to rice plants, food safety, and human health, thus the dynamic of Hg speciation in rice tissues was evaluated (Fig. 5). Due to transitional features, rice plants

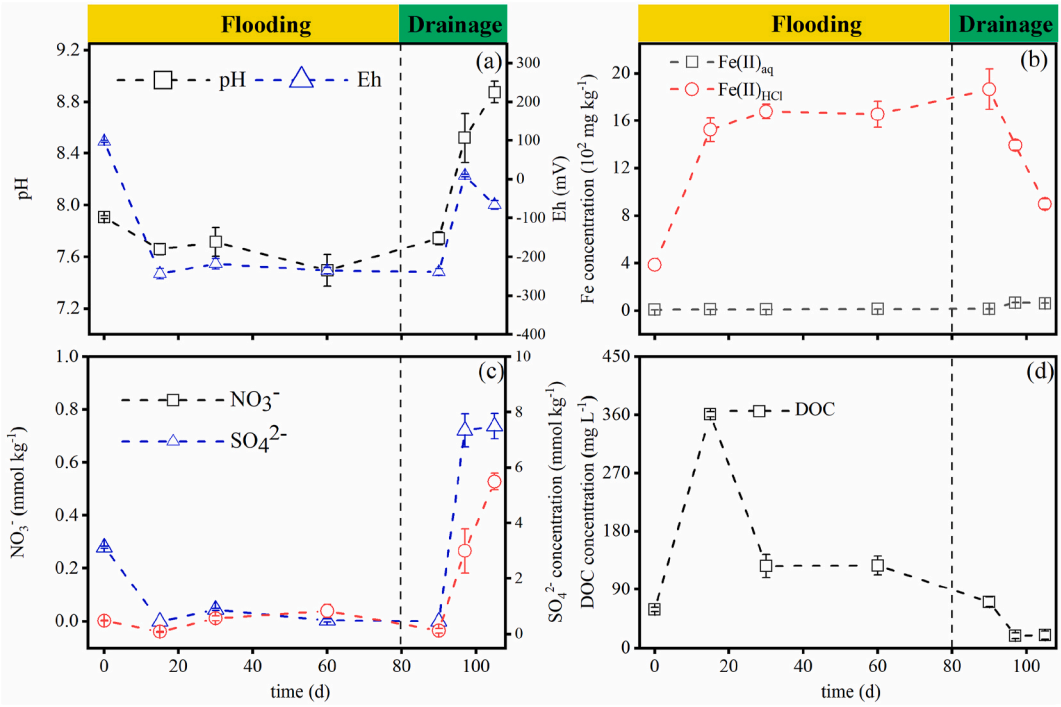


Fig. 3. (a) Paddy soil pH and Eh, (b) dissolved Fe(II) an $Fe(II)_{HCl}$ (c) NO_3^- and SO_4^{2-} (d) DOC concentrations in the process of rice growth.

uptake considerable Hg in soluble and exchangeable fraction from paddy soil to root dominantly through root hair surface contact, solute transport, and ion exchange during rice growth (Palansooriya et al., 2020). During flooding and drainage processes, the growth of rice took up soluble and exchangeable Hg fractions from paddy soil to rice tissues directly led to low levels of available IHg, WMeHg, and EMeHg in paddy soil (Fig. 4), increasing Hg ecotoxicological risk for rice (Liu et al., 2019). After adsorption of available Hg by rice, there are different proportions of IHg in rice tissues, and IHg contents was found to be highest in rice roots (Fig. 5a), indicating that rice roots have a strong ability to accumulate IHg. This was mainly due to the fact that bottom-up translocation of IHg from paddy soil to rice tissues would be blocked by endodermis casparian strips, causing stored Hg in the root bound to cell wall in rice plants (Yang et al., 2023a). During rice growth, IHg entered root cells from paddy soil and was transported upwards in the xylem, which required xylem loading by transporters (Wang et al., 2023a). Hg accumulation in rice is a result of enhanced expression of genes associated with GSH binding with Hg and Hg transport. Nowadays, genes shown to be related to Hg transport include *merA*, *merB*, and *merT*. Expression of *merT* or *merA* in Arabidopsis increased tolerance to Hg (Xu et al., 2017), and expression of *merB* alone enhanced Hg accumulation (Bizily et al., 1999). Other studies discovered that AtABCC1 and AtABCC2 are significant vacuolar transporters that mediate tolerance to, and transport of Hg (Park et al., 2012).

Specially, during the first 60 d of rice growth, the quantification of IHg contents in rice tissues followed the order: root > leaf > stem (Fig. 5a). After entering drainage, IHg levels in rice tissues followed the trend: root > leaf > stem > grain (Fig. 5a). Furthermore, IHg contents in rice root, stem, and leaf decreased over time, whereas IHg contents in rice grain increased with time (Fig. 5a). This implied that during rice growth, IHg could be absorbed by roots and then bottom-up translocation of IHg to the aerial part of rice plant, including rice root, stem, and leaf (Du et al., 2023; Meng et al., 2012). It has been reported that IHg levels in the leaves was controlled not only by paddy soil Hg concentration but also by atmospheric Hg. In the presence of Hg in the atmosphere, the translocation pathway of Hg in rice from atmospheric source to rice leaf, to the upper stalk, to the bottom stalk, and finally to rice root may also existed (Hu et al., 2023). However, no Hg was existed in the atmosphere during pot experiments, and thus, the absorption pathway of IHg and MeHg from air by rice leaves and then assimilation to the bottom stalk did not exist.

After 105 d of rice growth, IHg and MeHg levels in rice grain were 0.21 mg kg^{-1} and 33.44 μg kg^{-1}, respectively (Fig. 5b). Based on the National Food Hygiene Standard of China (GB 2762-2012), the maximum content of total Hg content was 0.02 mg kg^{-1} for vegetables (Hussain et al., 2022), and thus, IHg concentration in rice grain exceeded food safe limit. According to Joint FAO/WHO Expert Committee on Food Additives (JECFA) rule, the allowable intake of MeHg was 0.23 μg kg^{-1} of body weight (Joint, F.A.O.W.H.O.E.C.o.F.A., World Health, O., 2004). After calculation, a tolerable daily intake of MeHg in adults (based on a 60 kg body weight and daily intake of 500 g rice) was no more than 37.45 μg kg^{-1} body weight per day. This was slightly high MeHg contents in rice grain, indicating that the risk of MeHg in grain was relatively low.

3.5. Underlying mechanisms of Hg migration and transformation

In microcosm and pot experiments, most IHg, existed as organic- and sulfide-bound Hg and residual Hg, was relatively stable, and only a small portion of IHg was available during flooding and drainage (Figs. 2 and 4), which may be mainly because alkaline condition was beneficial to the tightly complexation of Hg and the componment of paddy soil. It has been reported that water regime and linked redox potential are significant factors influencing the variations of Hg species (Beckers et al., 2019a; Beckers et al., 2019b; Devai et al., 2005; Frohne et al., 2012; Wang et al., 2021b). After planting rice in a Hg-polluted alkaline paddy soil, water regime, redox potential, and geochemical of Fe, C, N, and S have changed with durations of flooding and drainage (Fig. 1), leading to residual Hg was dominant species with a small portion of available Hg (Fig. 4a), indicating that Hg mobility was very low during rice growth. During flooding, a small amount of IHg was released into solution due to a slight drop in pH (Fig. 1a) and reductive dissolution of Fe minerals (Fig. 2b). After entering drainage, an increase in pH (Fig. 1a and 3a) promoted the adsorption of Hg by paddy soil particles and newly-formed

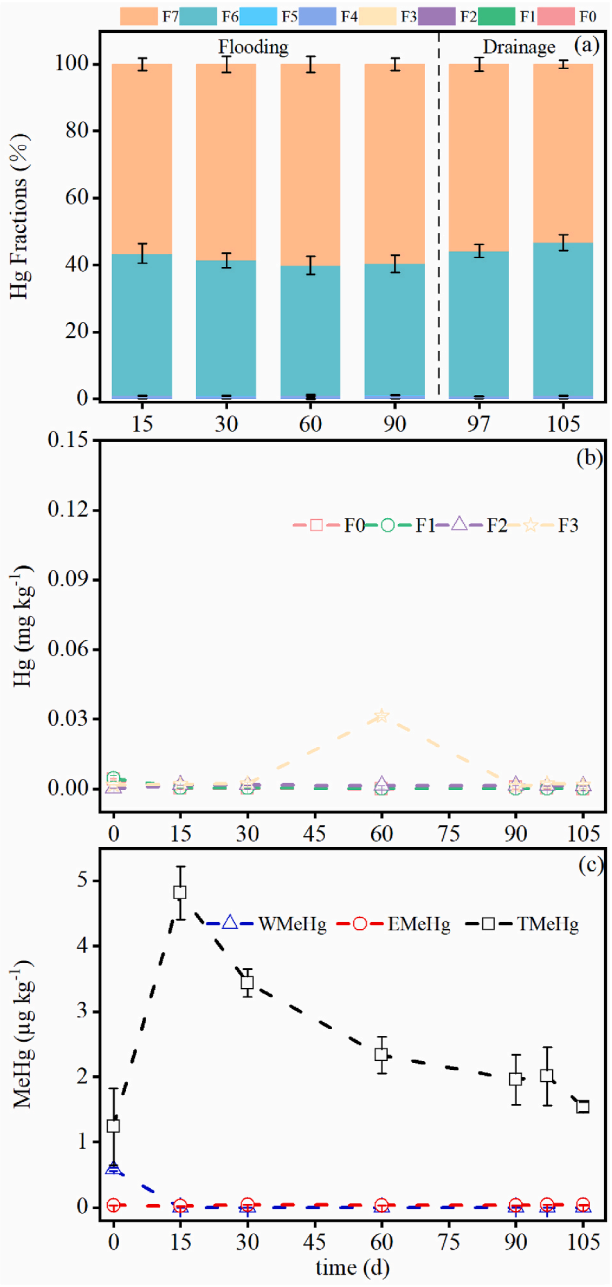


Fig. 4. Temporal changes of (a) Hg species (F0–F7) quantified by sequential chemical extraction, (b) available IHg concentration, and (c) TMeHg (total MeHg), WMeHg (water-soluble MeHg), and EMeHg (effective MeHg) contents. F0 denotes dissolved Hg; F1 denotes exchangeable Hg; F2 denotes carbonates bound Hg; F3 denotes fulvic acid bound Hg; F4 denotes humic acid bound Hg; F5 denotes Fe–Mn oxides bound Hg; F6 denotes organic and sulfides bound Hg; F7 denotes residual Hg.

Fe oxides from Fe(II) oxidation (Fig. 1b and 3b). Meanwhile, the oxidative decomposition of OM (Fig. 1d and 3d) and oxidative dissolution of HgS lead to Hg release. For the variations in MeHg, a minor fraction of IHg could be transformed into MeHg through Hg methylation during flooding (Fig. 2c and 4c), and under oxic condition, the demethylation of Hg reduced MeHg contents (Fig. 2c and 4c). The methylation of Hg under flooding condition was more favorable than that under drainage condition (Fig. 2c and 4c), which would aid in reducing MeHg accumulation in rice by shortening flooding time. During rice growth, IHg concentration was the highest in the root and the lowest in the grain (Fig. 5a). The IHg level in the grain was exceeded permitted concentration of standard, but MeHg concentration in the grain satisfied the human daily intake limit. Overall, planting rice in an Hg-polluted alkaline paddy soil was beneficial to decrease Hg environmental risk of at soil-water interfaces, but decreasing the safe rice production and consumption.

The knowledges obtained in this study are beneficial to an in-deep understanding of Hg geochemistry and hazards in an alkaline paddy soil. However, empirical or mechanistic models of Hg bioaccumulation and transport have not been established. The crucial research bottleneck is that the related genetic factors regulating MeHg or IHg accumulation in rice and the uptake and transport pathways of Hg remain unclear. Clearly, underling mechanisms of Hg methylation and demethylation in rice tissues are so complicated that they are poorly resolved. Uncovering the processes and potential mechanisms of Hg detoxification, transport, and uptake at molecular and cellular levels still needs considerable concern. Agricultural activities, including irrigation and fertilizer application, could introduce organics, cations, and anions into paddy soils, and change physicochemical characteristics and bacteria communities, leading to the variations in Hg transformation and mitigation (Hellal et al., 2015; Hu et al., 2023; O'Connor et al., 2019). Additionally, flooded rice fields are environments with strong variations of both environmental and management parameters thus influencing plant growth and pore water chemistry. Further investigations should be conducted to explore the effects of water management on MeHg production, which would promote our understanding of MeHg dynamics and to impede the uptake of Hg by rice and eventually realize safe rice production in a Hg-polluted alkaline paddy soil (Shen et al., 2021). In the future study, on the basis of identifying of the control genes of Hg uptake by rice, blocking the pathways/channels of Hg into rice cells will be an advantageous measure. It is worth to combined gene editing tool to develop novel rice varieties to impede Hg uptake by rice (Zafar et al., 2020).

4. Conclusions

Through the kinetics of microcosm and pot experiments, Hg environmental behavior and safe rice production in a contaminated alkaline paddy soil was investigated. At the interfaces of paddy soil and water, Hg transformation and mobility in microcosm experiment was similar to that in pot experiments. Specially, during flooding and drainage, quantitative results showed that available Hg content kept at low level and more than 98% Hg was organic and sulfide bound Hg and residual Hg. After planting rice in a polluted alkaline paddy soil, available Hg kept at low level. A small portion of IHg was transformed into MeHg during flooding, and the formed MeHg was converted back into IHg during drainage. Moreover, only a small amount of IHg and/or MeHg was uptake by rice during rice growth and the amount of IHg in the rice tissues gradually decreased. During rice growth, a portion of available Hg was transported from paddy soil to rice, leading to an environmental risk of Hg in rice grains. IHg levels in rice tissues followed the order: root > leaf > stem > grain. After rice ripening, IHg content in rice grain was exceeded safety threshold and consumption of rice posed a dietary health risk from MeHg exposure. These findings provided an in-depth insight into the biogeochemical cycling of Hg and Hg uptake by rice in an alkaline paddy soil.

CRediT authorship contribution statement

Shiwen Hu: Data curation, Funding acquisition, Investigation, Methodology, Visualization, Writing – original draft, Writing – review & editing. **Yufan Zhang:** Conceptualization, Data curation, Investigation, Methodology, Writing – review & editing. **Hanbing Meng:** Visualization, Writing – review & editing. **Yang Yang:** Writing – review & editing. **Guojun Chen:** Writing – review & editing. **Qi Wang:** Writing – review & editing. **Kuan Cheng:** Writing – review & editing. **Chao Guo:** Writing – review & editing. **Xiaomin Li:** Writing – review & editing.

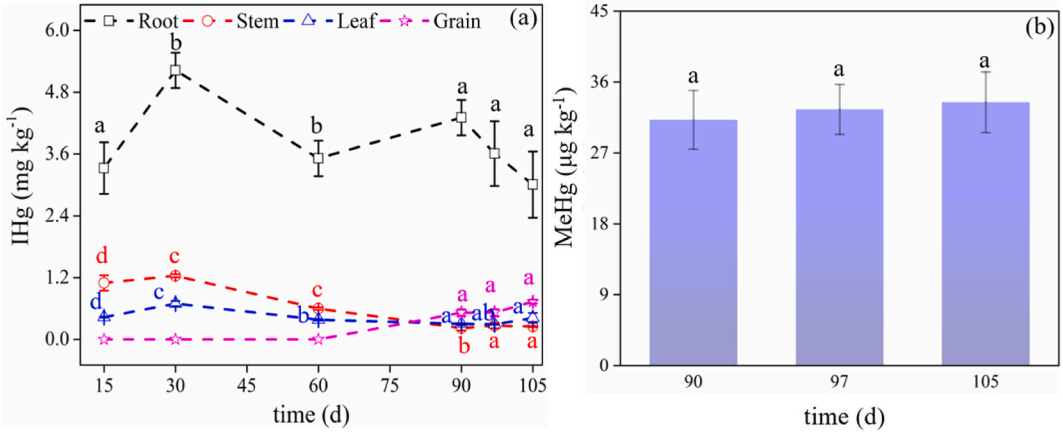


Fig. 5. (a) Temporal changes of inorganic mercury concentrations in various parts of rice plant (b) and MeHg in grains in the process of rice growth. Statistical discrepancies among different IHg concentrations were analyzed by Duncan's test ($p < 0.05$). Different letters in the columns denote a significant discrepancy at $p < 0.05$ using DMRT method.

Tongxu Liu: Conceptualization, Project administration, Supervision, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.envpol.2024.123471.

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